object.cnt

Purpose

`object.cnt` gives the basin name, the basin areas and the number of spatial objects of each type: HRUs, routing units, aquifers, channels, reservoirs, point sources and the others listed below. SWAT+ uses the counts to set up its list of objects before any connection file is read. The file is named in slot 4 of the `simulation` line of `file.cio` (default `object.cnt`). It is required: if it is missing or `null`, SWAT+ prints `Cannot find object.cnt input file` and stops.

Format

Free format, whitespace-separated. Line 1 is a title and line 2 is the column header; both are skipped. Line 3 contains 21 values in the order below. The reader does not match columns by header name.

#	Column	Units	Default	Description
1	`name`	-	blank	Basin name, up to 25 characters. It appears in the name column of the basin output files.
2	`ls_area`	ha	0	Land area of the basin.
3	`tot_area`	ha	0	Total area of the basin, including reservoirs and other water bodies.
4	`obj`	count	0	Total number of spatial objects. Must equal the sum of columns 5 to 18.
5	`hru`	count	0	Number of HRUs (`hru.con`).
6	`lhru`	count	0	Number of HRU-lte objects (`hru-lte.con`).
7	`rtu`	count	0	Number of routing units (`rout_unit.con`).
8	`mfl`	count	0	Number of gwflow objects (`gwflow.con`). See gwflow.
9	`aqu`	count	0	Number of aquifers (`aquifer.con`).
10	`cha`	count	0	Number of legacy channels (`channel.con`).
11	`res`	count	0	Number of reservoirs (`reservoir.con`).
12	`rec`	count	0	Number of recall objects: point sources and inlets (`recall.con`).
13	`exco`	count	0	Number of export coefficient objects (`exco.con`).
14	`dlr`	count	0	Number of delivery ratio objects (`delratio.con`).
15	`can`	count	0	Number of canals. No connection file is read for canals in 62.0.1.
16	`pmp`	count	0	Number of pumps. No connection file is read for pumps in 62.0.1.
17	`out`	count	0	Number of outlets (`outlet.con`).
18	`lcha`	count	0	Number of channels routed with the SWAT+ channel method (`chandeg.con`).
19	`aqu2d`	count	0	Number of 2D aquifers. Read and not used.
20	`hrd`	count	0	Number of herds. Read and not used.
21	`wro`	count	0	Number of water rights objects. Read and not used.

The column headers are the ones the SWAT+ Editor writes. The old GitBook documentation calls column 8 `gwfl`. Only the order matters to SWAT+.

Object counts

Each count must match the number of objects in the corresponding connection file named on the `connect` line of `file.cio`. SWAT+ numbers all objects in one list, in this order: HRUs, HRU-lte objects, routing units, gwflow objects, aquifers, legacy channels, reservoirs, recall objects, export coefficient objects, delivery ratio objects, canals, pumps, outlets and SWAT+ channels. The object list has `obj` entries, so `obj` must equal the sum of these counts. Counts for canals and pumps reserve places in the list, but no objects are read into them.

If line 3 ends early, the reader reaches the end of the file and the missing counts stay at zero. The `refdata/Osu_1hru` example uses this and leaves out `hrd` and `wro`. If the file has more lines after line 3, SWAT+ keeps reading them in groups of title, header and values, and the last complete group is the one used. A file with a single data line avoids this.

Basin areas

`ls_area` and `tot_area` are not used to compute HRU or object areas; those come from the connection files. They are used in these places:

- `ls_area`: the HRU area when a decision table changes the fraction of the basin an HRU covers, and the HRU areas of the regions in soft calibration (`water_balance.sft`, see soft calibration).
- `tot_area`: the area weights for the basin aquifer output (`basin_aqu_*` files), and the conversion of gwflow basin water balance volumes to depths.

gwflow

When `gwflow = 1` in `codes.bsn`, SWAT+ looks for `chancell.gw` in the run folder after reading `object.cnt`:

- If `chancell.gw` exists and `mfl` is 0, SWAT+ counts the channel cell rows in `chancell.gw` and uses that count as the number of gwflow objects. It adds that count to `obj`, removes the aquifers from `obj`, sets the aquifer count to 0 and reads the gwflow connections from `gwflow.con` in place of `aquifer.con`.
- If `chancell.gw` exists and `mfl` is already above 0, SWAT+ keeps `mfl` and `obj` as given and sets the aquifer count to 0. In that case `obj` should already leave out the aquifers.
- If `chancell.gw` is missing or has no data after its header, SWAT+ turns gwflow off and the run uses the lumped aquifers.

In all three cases SWAT+ opens the `gwflow_record` log file. See gwflow for the gwflow input files.

Example

`refdata/Ames_sub1/object.cnt` in the SWAT+ repository:

```
object.cnt:
name                    ls_area      tot_area       obj       hru      lhru       rtu       mfl       aqu
demo                       1.           1            12         12         0         0         0         0
```

The basin is named `demo` and both areas are 1 ha. The project has 12 objects, all of them HRUs, so `obj` equals `hru`. There are no routing units, aquifers or channels.

`refdata/Osu_1hru/object.cnt` has 4 objects: one HRU, one routing unit, one aquifer and one SWAT+ channel (`lcha`). Its data line stops after `aqu2d`, and the `hrd` and `wro` counts are taken as zero.

Related files

- `file.cio`: names this file and the connection files.
- Connection files: one row per object; the number of rows must match the counts here.
- `codes.bsn`: its `gwflow` code starts the gwflow check above.
- gwflow: `chancell.gw` and the other gwflow inputs.

Sources

- Reader: basin_read_objs.f90, called from proc_bsn.f90 after `codes.bsn`
- Type definitions: `type basin_inputs` in basin_module.f90, `type spatial_objects` in hydrograph_module.f90
- Object numbering and connection files: hyd_connect.f90
- Uses of the areas: hru_fr_change.f90, lcu_read_softcal.f90, basin_aquifer_output.f90, gwflow_output.f90
- File name slot: `in_sim%object_cnt` in input_file_module.f90

Column	Fortran variable
name	bsn%name
ls_area	bsn%area_ls_ha
tot_area	bsn%area_tot_ha
obj	sp_ob%objs
hru	sp_ob%hru
lhru	sp_ob%hru_lte
rtu	sp_ob%ru
mfl	sp_ob%gwflow
aqu	sp_ob%aqu
cha	sp_ob%chan
res	sp_ob%res
rec	sp_ob%recall
exco	sp_ob%exco
dlr	sp_ob%dr
can	sp_ob%canal
pmp	sp_ob%pump
out	sp_ob%outlet
lcha	sp_ob%chandeg
aqu2d	sp_ob%aqu2d
hrd	sp_ob%herd
wro	sp_ob%wro